

CHAPTER 4

Charged Particle Motion in Time-varying Electromagnetic Fields

1. INTRODUCTION

In this chapter we analyze the motion of charged particles in the presence of time-varying fields. Initially, in the two following sections, we consider a time-varying electric field and a constant magnetic field, both fields being spatially uniform. The assumption of a constant and spatially uniform $\vec{B}$ field is well justified if the externally applied magnetostatic field is much larger than the magnetic field associated with the time-varying $\vec{E}$ field. Also, the assumption of an electric field which is spatially uniform is valid if the charged particle cyclotron radius is much smaller than the scale length of the spatial variation of $\vec{E}$. Both these requirements are assumed to hold in the analysis presented in sections 2 and 3. In section 4 we consider a time-varying magnetic field and the corresponding space-varying electric field.

2. SLOWLY TIME-VARYING ELECTRIC FIELD

2.1 Equation of motion and polarization drift

For the moment we shall assume that the characteristic time scale of variation of the electric field is much larger than the particle cyclotron period. The component of the motion of the charged particle along the magnetic field lines is given by (2.5.4), from which we can write in general

$$\vec{v}_{\parallel}(t) - \vec{v}_{\parallel}(0) = \frac{q}{m} \int_0^t \vec{E}_{\parallel}(t') dt' \quad (2.1)$$

This result however does not lead to any new interesting information.

Since the $\vec{E}$ field is varying slowly in time, the component of the motion across the magnetic field lines is expected to be not very different from

that for a constant $\vec{E}$ field. Therefore, it is reasonable to seek the solution for $\vec{v}_\perp$ in a form similar to (2.5.8). Hence, we take

$$\vec{v}_\perp = \vec{v}_\perp' + \vec{v}_E + \vec{v}_p \quad (2.2)$$

where $\vec{v}_E = \vec{E}_\perp \times \vec{B}/B^2$ is the electromagnetic plasma drift velocity (2.5.11). Note that $\vec{v}_E$ varies slowly with time, since $\vec{E}$ is slowly time-varying. The substitution of (2.2) into the perpendicular component equation of motion (2.5.5) yields

$$m \frac{d}{dt} (\vec{v}_\perp' + \vec{v}_E + \vec{v}_p) = q [\vec{E}_\perp + (\vec{v}_\perp' + \vec{v}_E + \vec{v}_p) \times \vec{B}] \quad (2.3)$$

Using (2.5.9) we can write (2.3) as

$$m \frac{d\vec{v}_\perp'}{dt} + m \frac{d}{dt} \left(\frac{\vec{E}_\perp \times \vec{B}}{B^2} \right) + m \frac{d\vec{v}_p}{dt} = q \vec{v}_\perp' \times \vec{B} + q \vec{v}_p \times \vec{B} \quad (2.4)$$

Thus, if we set

$$\vec{v}_p = (m/q)(\partial \vec{E}_\perp / \partial t)/B^2 \quad (2.5)$$

(2.4) reduces to

$$m d\vec{v}_\perp'/dt + m d\vec{v}_p/dt = q \vec{v}_\perp' \times \vec{B} \quad (2.6)$$

If the second term on the left-hand side can be neglected, then this equation becomes identical to (2.5.12), which describes a circular motion about the magnetic field lines. Comparing the relative magnitudes of the second term on the left, with that on the right-hand side of (2.6), we find

$$\begin{aligned} \frac{|m d\vec{v}_p/dt|}{|q \vec{v}_\perp' \times \vec{B}|} &= \frac{|m^2 (\partial^2 \vec{E}_\perp / \partial t^2) / q B^2|}{|q B v_\perp'|} = \left| \frac{(E_\perp / B)}{v_\perp'} \left(\frac{\omega^2 m^2}{q^2 B^2} \right) \right| = \\ &= |v_E / v_\perp'| (\omega / \omega_c)^2 \end{aligned} \quad (2.7)$$

where we have assumed that $\vec{E}_\perp$ has a harmonic time dependence with a characteristic angular frequency ω . Considering that this characteristic frequency is much smaller than the cyclotron frequency,

$$\omega \ll \omega_c \quad (2.8)$$

and, further, if $|\vec{v}_E/\vec{v}_\perp|$ is also small, then the term $m d\vec{v}_p/dt$ can be neglected in comparison with the other terms of (2.6), and we obtain

$$m d\vec{v}_\perp/dt = q \vec{v}_\perp \times \vec{B} \quad (2.9)$$

which is identical to (2.5.12). Therefore, $\vec{v}_\perp$ corresponds to the usual circular motion of the charged particle about the magnetic field, and is independent of the variations of the electric field. Superposed upon this circular motion are the drift velocities

$$\vec{v}_E = (\vec{E}_\perp \times \vec{B})/B^2 \quad (2.10)$$

and

$$\vec{v}_p = m(\partial \vec{E}_\perp / \partial t) / (q B^2) \quad (2.11)$$

Thus, the effect of a slowly varying electric field is the addition of the velocity $\vec{v}_p$ called the *polarization drift velocity*.

Since $\vec{v}_p$ is in opposite directions for charges of opposite sign, the time dependent electric field produces a net polarization current in a neutral plasma, so that the plasma medium behaves like a dielectric. The *polarization current density* $\vec{J}_p$ is the rate of flow of positive and negative charges across unit area, and is given by

$$\vec{J}_p = (1/\Delta V) \sum_i q_i \vec{v}_{pi} = [(1/\Delta V) \sum_i m_i] (\partial \vec{E}_\perp / \partial t) / B^2 = \rho_m (\partial \vec{E}_\perp / \partial t) / B^2 \quad (2.12)$$

where the summation is over all positive and negative charges contained in the small volume element ΔV , and ρ_m is the mass density of the plasma.

2.2 Plasma dielectric constant

The polarization effect in a plasma is due to the time variation of the electric field. The application of a steady $\vec{E}$ field does not result in a polarization field, since the ions and electrons can move around to preserve quasineutrality. Since the plasma behaves like a dielectric, the polarization current density $\vec{J}_p$ can be taken into account through the introduction of the

dielectric constant of the plasma. For this purpose, we can separate the *total* current density $\vec{J}$ into the polarization current density $\vec{J}_p$ and the current density $\vec{J}_0$ due to other sources,

$$\vec{J} = \vec{J}_p + \vec{J}_0 \quad (2.13)$$

Thus, combining $\vec{J}_p$ with the term $\epsilon_0 \partial \vec{E}_\perp / \partial t$ which appears on the right-hand side of Maxwell $\vec{\nabla} \times \vec{B}$ equation, we obtain

$$\epsilon_0 (\partial \vec{E}_\perp / \partial t) + (\rho_m / B^2) (\partial \vec{E}_\perp / \partial t) = \epsilon_0 (1 + \rho_m / \epsilon_0 B^2) \partial \vec{E}_\perp / \partial t = \epsilon (\partial \vec{E}_\perp / \partial t) \quad (2.14)$$

where

$$\epsilon = \epsilon_0 \epsilon_r = \epsilon_0 (1 + \rho_m / \epsilon_0 B^2) \quad (2.15)$$

is the *effective electric permittivity* perpendicular to the magnetic field. For low frequencies, the *relative permittivity* ϵ_r of a plasma can be very high; for a number density of 10^{20} particles/m³ and $B = 1$ Tesla we have $\epsilon_r \approx 10^4$.

The resulting charge density ρ_p , which accumulates as a result of the polarization current density $\vec{J}_p$, satisfies the charge continuity equation

$$\partial \rho_p / \partial t + \vec{\nabla} \cdot \vec{J}_p = 0 \quad (2.16)$$

From (2.16) and (2.12) we obtain

$$\rho_p = -(\rho_m / B^2) \vec{\nabla} \cdot \vec{E}_\perp \quad (2.17)$$

The total charge density ρ can be separated as

$$\rho = \rho_0 + \rho_p \quad (2.18)$$

where ρ_0 corresponds to $\vec{J}_0$. Assuming that the parallel component of the electric field vanishes, we see that

$$\vec{\nabla} \cdot \vec{E} = (\rho_0 + \rho_p) / \epsilon_0 = \rho_0 / \epsilon_0 - (\rho_m / \epsilon_0 B^2) \vec{\nabla} \cdot \vec{E} \quad (2.19)$$

from which we find, using (2.15),

$$\vec{\nabla} \cdot \vec{E} = \rho_0 / \epsilon \quad (2.20)$$

Thus, the resulting charge density ρ_p can also be correctly taken into account by the introduction of an effective electric permittivity ϵ .

We can further verify the correctness of introducing the effective electric permittivity of the plasma, by calculating the *total energy density* associated with the $\vec{E}$ field, which for an ordinary dielectric medium of effective permittivity ϵ is given by $\epsilon E^2/2$. The *energy density* in the electric field is given by

$$W_E = \epsilon_0 E^2/2 \quad (2.21)$$

To calculate the additional drift kinetic energy acquired by the particle as a result of the polarization drift, we note that for a change $\Delta \vec{E}_1$ in the electric field, in a time interval Δt , the displacement $\Delta \vec{r}$ of the guiding center due to this change, is

$$\Delta \vec{r} = \vec{v}_p \Delta t = (m/qB^2)(\partial \vec{E}_1 / \partial t) \Delta t = (m/qB^2) \Delta \vec{E}_1 \quad (2.22)$$

The corresponding work done by the electric field is, using (2.22),

$$\Delta W = q \vec{E}_1 \cdot \Delta \vec{r} = (m/B^2) \vec{E}_1 \cdot \Delta \vec{E}_1 = \Delta(mE_1^2/2B^2) \quad (2.23)$$

Hence, using (2.10), the change in the kinetic energy of the particle associated with the polarization drift is given by

$$\Delta W = \Delta(mv_E^2/2) \quad (2.24)$$

This result shows that the work done by the electric field, during polarization, is equal to the change in the kinetic energy associated with the particle motion at the electromagnetic drift velocity $\vec{v}_E$. Note that $\vec{v}_E$ does not lead to any energy exchange between field and particle, since the displacement associated with $\vec{v}_E$ is perpendicular to the electric field. Summing (2.24) over all particles in a unit volume gives the change in the total kinetic energy density of the system

$$\Delta W_V = \Delta(\rho_m v_E^2/2) = \Delta(\rho_m E_1^2/2B^2) \quad (2.25)$$

The kinetic energy density associated with the circular motion of the particles is not affected by changes in the electric field. Thus, the total energy density associated with the electric field is

$$\begin{aligned} W_E + W_V &= \epsilon_0 E^2/2 + \rho_m v^2/2 \\ &= (\epsilon_0 E^2/2)(1 + \rho_m/\epsilon_0 B^2) = \epsilon E^2/2 \end{aligned} \quad (2.26)$$

assuming that there is no parallel component of the electric field. This result completes our discussion about the legitimacy of the introduction of an effective electric permittivity for the plasma.

3. ELECTRIC FIELD WITH ARBITRARY TIME VARIATION

3.1 Solution of the equation of motion

We consider now an arbitrary time variation of the electric field but, again, the field is uniform in space. The applied magnetic field is static and uniform, as before. Without loss of generality, the time variation of $\vec{E}$ can be assumed to be harmonic with angular frequency ω ,

$$\vec{E}(t) = \vec{E}_0 e^{-i\omega t} \quad (3.1)$$

where the complex amplitude $\vec{E}_0$ is independent of time. According to the usual convention, only the real part of this expression is to be taken. An arbitrary time variation of $\vec{E}$ can be written as a superposition of terms similar to (3.1), corresponding to all possible values of ω , since the equation of motion (2.5.5) is linear. Using (3.1), the equation of motion becomes

$$m \, d\vec{v}/dt = q (\vec{E}_0 e^{-i\omega t} + \vec{v} \times \vec{B}) \quad (3.2)$$

It is natural to expect the forced oscillations of the charged particles to have the same frequency as that of the forcing electric field. Thus, the particle velocity vector may be conveniently decomposed into two parts

$$\vec{v} = \vec{v}_m + \vec{v}_e e^{-i\omega t} \quad (3.3)$$

where $\vec{v}_m$ is the velocity associated with the magnetic field alone and, thus, contains no time variation at the angular frequency ω , while $\vec{v}_e$ is due to the oscillating electric field. The substitution of (3.3) into (3.2) gives

$$m(d\vec{v}_m/dt - i\omega\vec{v}_e e^{-i\omega t}) = q [\vec{E}_0 e^{i\omega t} + \vec{v}_m \times \vec{B} + (\vec{v}_e \times \vec{B}) e^{-i\omega t}] \quad (3.4)$$

The terms containing the periodicity with the angular frequency ω are decoupled from those that do not, so that (3.4) separates into two equations,

$$m d\vec{v}_m/dt = q \vec{v}_m \times \vec{B} \quad (3.5)$$

$$- m i\omega\vec{v}_e = q (\vec{E}_0 + \vec{v}_e \times \vec{B}) \quad (3.6)$$

Equation (3.5) corresponds to the usual circular motion of the particle about the magnetic field lines at the cyclotron frequency ω_c .

To solve (3.6) for $\vec{v}_e$ it is appropriate to separate it into components parallel and perpendicular to $\vec{B}$. The *parallel* velocity component is obtained immediately as

$$\vec{v}_{e\parallel} = (i/\omega)(q/m) \vec{E}_{0\parallel} \quad (3.7)$$

while the *perpendicular* velocity component satisfies the equation

$$[-i\omega + (q/m) \vec{B} \times] \vec{v}_{e\perp} = (q/m) \vec{E}_{0\perp} \quad (3.8)$$

Introducing the cyclotron frequency vector

$$\vec{\omega}_c = - q\vec{B}/m \quad (3.9)$$

we can rewrite (3.8) in the form

$$(i\omega + \vec{\omega}_c \times) \vec{v}_{e\perp} = - (q/m) \vec{E}_{0\perp} \quad (3.10)$$

In order to solve this equation for $\vec{v}_{e\perp}$, we multiply both sides by the conjugate operator $-(i\omega - \vec{\omega}_c \times)$. First, we note that

$$\begin{aligned}
 (i\omega - \vec{\omega}_c \times)(i\omega + \vec{\omega}_c \times) \vec{v}_{e\perp} &= (i\omega)^2 \vec{v}_{e\perp} - \vec{\omega}_c \times (\vec{\omega}_c \times \vec{v}_{e\perp}) \\
 &= (\omega_c^2 - \omega^2) \vec{v}_{e\perp}
 \end{aligned} \tag{3.11}$$

Therefore, (3.10) becomes

$$\vec{v}_{e\perp} = (q/m) (i\omega - \vec{\omega}_c \times) \vec{E}_{0\perp} / (\omega^2 - \omega_c^2) \tag{3.12}$$

Combining the results contained in (3.12), (3.7) and (3.5), we obtain the following expression for the total velocity vector (3.3)

$$\vec{v} = \vec{v}_m + (q/m) [(i/\omega) \vec{E}_{0\parallel} + (i\omega - \vec{\omega}_c \times) \vec{E}_{0\perp} / (\omega^2 - \omega_c^2)] e^{-i\omega t} \tag{3.13}$$

3.2 Physical interpretation

The result contained in (3.13) shows that *along the magnetic field lines* the particle oscillates with frequency ω and amplitude $\vec{v}_{e\parallel}$, as given by (3.7), so that the velocity oscillation lags 90 degrees behind the oscillation of the applied electric field. This result is easily seen by taking the real part of the vectors $\vec{v}_{e\parallel} e^{-i\omega t}$ and $\vec{E}_{0\parallel} e^{-i\omega t}$

$$\text{Re} \{ \vec{v}_{e\parallel} e^{-i\omega t} \} = (q/\omega m) \vec{E}_{0\parallel} \sin(\omega t) \tag{3.14}$$

$$\text{Re} \{ \vec{E}_{0\parallel} e^{-i\omega t} \} = \vec{E}_{0\parallel} \cos(\omega t) \tag{3.15}$$

which clearly are 90 degrees out of phase. In the *plane perpendicular to $\vec{B}$* the particle motion is the superposition of the circular motion at the cyclotron frequency ω_c , with an oscillation at the frequency ω and amplitude given by (3.12).

In order to analyze the physical meaning of the motion in the *plane perpendicular to $\vec{B}$* , it is convenient to decompose the oscillating electric field vector into two circularly polarized components, with opposite directions of rotation. The advantage of using the two circularly polarized components is that (3.12), for the perpendicular velocity component $\vec{v}_{e\perp}$, uncouples into two separate equations pertaining to the two circular polarizations rotating in opposite directions. Thus, we take

$$\vec{E}_\perp = \vec{E}_R + \vec{E}_L = (\vec{E}_{0R} + \vec{E}_{0L}) e^{-i\omega t} \tag{3.16}$$

with

$$\vec{E}_{OR} = (\vec{E}_{O\perp} + i \vec{B} \times \vec{E}_{O\perp})/2 \quad (3.17)$$

$$\vec{E}_{OL} = (\vec{E}_{O\perp} - i \vec{B} \times \vec{E}_{O\perp})/2 \quad (3.18)$$

where $\vec{B} = \vec{B}/B$ is a unit vector pointing in the direction of the magnetic field. The component $\vec{E}_R$ represents a circularly polarized field with its electric vector rotating to the *right* (*clockwise* direction), and the component $\vec{E}_L$ represents a circularly polarized field with its electric vector rotating to the *left* (*counterclockwise* direction), as seen by an observer looking in the direction of the $\vec{B}$ field.

To understand the physical meaning of this decomposition, let us consider a Cartesian coordinate system with the z axis pointing along $\vec{B}$ and the x axis pointing along $\vec{E}_{O\perp}$. Then, we have

$$\vec{B} \times \vec{E}_{O\perp} = \hat{z} \times (\hat{x} E_{O\perp}) = E_{O\perp} \hat{y} \quad (3.19)$$

and from (3.17) and (3.18),

$$\vec{E}_R = \vec{E}_{OR} e^{-i\omega t} = (E_{O\perp}/2)(\hat{x} + i\hat{y}) e^{-i\omega t} \quad (3.20)$$

$$\vec{E}_L = \vec{E}_{OL} e^{-i\omega t} = (E_{O\perp}/2)(\hat{x} - i\hat{y}) e^{-i\omega t} \quad (3.21)$$

The components of the actual electric field are obtained by taking the real part of these two equations,

$$\text{Re } \{E_R\} = (E_{O\perp}/2)(\hat{x} \cos\omega t + \hat{y} \sin\omega t) \quad (3.22)$$

$$\text{Re } \{E_L\} = (E_{O\perp}/2)(\hat{x} \cos\omega t - \hat{y} \sin\omega t) \quad (3.23)$$

Thus, the fields $\vec{E}_R$ and $\vec{E}_L$ are constant in magnitude, but sweep around in a circle at frequency ω . For $\vec{E}_R$ the rotation is clockwise looking in the direction of $\vec{B}$, and it is called a *right circularly polarized* component, while for $\vec{E}_L$ the rotation is counterclockwise and it is called a *left circularly polarized* component. This is illustrated in Fig. 1.

Proceeding in the analysis of the particle motion in the perpendicular plane, we must now substitute the decomposed form (3.16) into (3.12). The operator $(i\omega - \vec{\omega}_c \times)$ when applied on the right circularly polarized component $\vec{E}_{oR}$ gives

$$\begin{aligned}
 (i\omega - \vec{\omega}_c \times) \vec{E}_{oR} &= (i\omega - \vec{\omega}_c \times)(\vec{E}_{o\perp} + i \mathbf{B} \times \vec{E}_{o\perp})/2 \\
 &= [i\omega (\vec{E}_{o\perp} + i \mathbf{B} \times \vec{E}_{o\perp}) + (qB/m) \mathbf{B} \times \vec{E}_{o\perp} - i(qB/m) \vec{E}_{o\perp}]/2 \\
 &= [i\omega (\vec{E}_{o\perp} + i \mathbf{B} \times \vec{E}_{o\perp}) - i(qB/m) (\vec{E}_{o\perp} + i \mathbf{B} \times \vec{E}_{o\perp})]/2 \\
 &= i(\omega - qB/m) \vec{E}_{oR}
 \end{aligned} \tag{3.24}$$

In a similar fashion we find for the left circularly polarized component,

$$(i\omega - \vec{\omega}_c \times) \vec{E}_{oL} = i(\omega + qB/m) \vec{E}_{oL} \tag{3.25}$$

Therefore, both $\vec{E}_{oR}$ and $\vec{E}_{oL}$ are eigenvectors of the complex operator appearing in (3.12). Using (3.24) and (3.25), (3.12) becomes

$$\vec{v}_{e\perp} = i(q/m) \left[\frac{\vec{E}_{oR}}{(\pm \omega_c + \omega)} - \frac{\vec{E}_{oL}}{(\pm \omega_c - \omega)} \right] \tag{3.26}$$

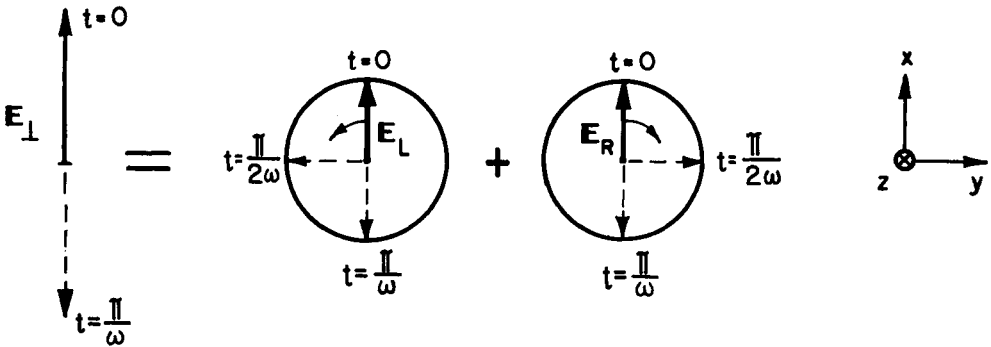


Fig. 1 - A plane polarized wave, indicated by the double-headed arrow $\vec{E}_{\perp}$, is equivalent to the sum of left and right circularly polarized waves $\vec{E}_L$ and $\vec{E}_R$.

The upper sign applies to a positively charged particle and the lower sign to a negatively charged particle, since $\omega_c = |q| B/m$ is always positive. Thus, $\vec{v}_{e\perp}$ also separates into two vectors rotating in opposite directions,

$$\vec{v}_{e\perp} = \vec{v}_R + \vec{v}_L \quad (3.27)$$

where

$$\vec{v}_R = i(q/m) \frac{\vec{E}_{oR}}{(\pm \omega_c + \omega)} \quad (3.28)$$

$$\vec{v}_L = -i(q/m) \frac{\vec{E}_{oL}}{(\pm \omega_c - \omega)} \quad (3.29)$$

For a *positive ion* we see that, as ω approaches the ion cyclotron frequency ω_{ci} , there is resonance between the ion and the *left* circularly polarized component of the $\vec{E}$ field. For an *electron*, as ω approaches the electron cyclotron frequency ω_{ce} , resonance occurs with the *right* circularly polarized component.

3.3 Mobility dyad

The expression for $\vec{v}_e = \vec{v}_{e''} + \vec{v}_R + \vec{v}_L$ can be written in a compact form through the introduction of the *mobility dyad* $\bar{\bar{\mu}}$, defined by

$$\vec{v}_e = \bar{\bar{\mu}} \cdot \vec{E}_0 \quad (3.30)$$

Using (3.7), (3.28) and (3.29) we see that the mobility dyad $\bar{\bar{\mu}}$ is diagonal in the rotating system, so that (3.30) becomes in matrix form,

$$\begin{pmatrix} v_R \\ v_L \\ v_{e''} \end{pmatrix} = \frac{i}{m} \frac{q}{\omega} \begin{pmatrix} \frac{\omega}{(\pm \omega_c + \omega)} & 0 & 0 \\ 0 & \frac{-\omega}{(\pm \omega_c - \omega)} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} E_{oR} \\ E_{oL} \\ E_{o''} \end{pmatrix} \quad (3.31)$$

If, instead of the rotating system, we use a stationary Cartesian coordinate system with the z axis pointing along the magnetostatic field, we can write (3.7) and (3.12) in matrix form as

$$\begin{pmatrix} v_{ex} \\ v_{ey} \\ v_{ez} \end{pmatrix} = \frac{i}{m} \frac{q}{\omega} \begin{pmatrix} \frac{\omega^2}{(\omega^2 - \omega_c^2)} & \frac{\pm i \omega_c \omega}{(\omega^2 - \omega_c^2)} & 0 \\ -\frac{\pm i \omega_c \omega}{(\omega^2 - \omega_c^2)} & \frac{\omega^2}{(\omega^2 - \omega_c^2)} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} E_{ox} \\ E_{oy} \\ E_{oz} \end{pmatrix} \quad (3.32)$$

3.4 Plasma conductivity dyad

Denoting by n_0 the number density of the electrons (charge - e) and ions (charge + e) in a plasma, the electric current density is given by

$$\vec{J} = -n_0 e \vec{v}_e + n_0 e \vec{v}_i = -n_0 e \bar{\mu}_e \cdot \vec{E} + n_0 e \bar{\mu}_i \cdot \vec{E} \quad (3.33)$$

where $\bar{\mu}_e$ and $\bar{\mu}_i$ are the mobility dyads for the electrons and ions, respectively. Introducing the conductivity dyad $\bar{\sigma}$ by

$$\vec{J} = \bar{\sigma} \cdot \vec{E} = (\bar{\sigma}_e + \bar{\sigma}_i) \cdot \vec{E} \quad (3.34)$$

we obtain from (3.33), for the electron and ion conductivities, respectively,

$$\bar{\sigma}_e = -n_0 e \bar{\mu}_e \quad (3.35)$$

$$\bar{\sigma}_i = n_0 e \bar{\mu}_i \quad (3.36)$$

With the help of (3.32), these can be expressed in matrix form as

$$\bar{\sigma}_e = \frac{i n_0 e^2}{m_e \omega} \begin{pmatrix} \frac{\omega^2}{(\omega^2 - \omega_{ce}^2)} & \frac{-i \omega_{ce} \omega}{(\omega^2 - \omega_{ce}^2)} & 0 \\ \frac{i \omega_{ce} \omega}{(\omega^2 - \omega_{ce}^2)} & \frac{\omega^2}{(\omega^2 - \omega_{ce}^2)} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (3.37)$$

$$\bar{\sigma}_i = \frac{i n_0 e^2}{m_i \omega} \begin{pmatrix} \frac{\omega^2}{(\omega^2 - \omega_{ci}^2)} & \frac{i \omega_{ci} \omega}{(\omega^2 - \omega_{ci}^2)} & 0 \\ \frac{-i \omega_{ci} \omega}{(\omega^2 - \omega_{ci}^2)} & \frac{\omega^2}{(\omega^2 - \omega_{ci}^2)} & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (3.38)$$

The fact that the conductivity $\bar{\sigma}$ is imaginary implies that $\vec{J}$ and $\vec{E}$ are 90° out of phase, since the actual physical expressions for $\vec{J}$ and $\vec{E}$ are obtained by taking the real part of (3.34) and (3.1), respectively.

3.5 Cyclotron resonance

The particle velocity, given in (3.7), (3.28) and (3.29), does not represent correctly the motion of the particle when the frequency ω of the electric field is equal to the particle cyclotron frequency ω_c . A positive particle, for example, when the forcing electric field rotates in the counterclockwise direction looking in the direction of $\vec{B}$ (left circularly polarized component), is able to absorb energy from the electric field, so that its speed increases continually and indefinitely with time. The same is true for an electron, say, and the right circularly polarized component of the forcing electric field when $\omega = \omega_{ce}$. This phenomenon is called *cyclotron resonance*. To investigate the particle motion under resonance conditions, it is necessary to go back to the original equations of motion and solve the problem for the case when $\omega = \omega_c$.

For simplicity, let us assume that the component of the $\vec{E}$ field along $\vec{B}$ vanishes, that is, $\vec{E}_\parallel = 0$. Hence, we take

$$\vec{E}(t) = \vec{E}_{0\perp} e^{-i\omega_c t} \quad (3.39)$$

with $\omega_c = |q| B/m$. In view of this assumption, the particle velocity *along* $\vec{B}$ is constant and is equal to the initial parallel velocity. The component of the equation of motion in the plane *normal* to the magnetic field becomes

$$d\vec{v}_\perp(t)/dt = (q/m) [\vec{E}_{0\perp} e^{-i\omega_c t} + \vec{v}_\perp(t) \times \vec{B}] \quad (3.40)$$

Differentiating this equation with respect to time, yields

$$d^2\vec{v}_\perp/dt^2 = (q/m) [-i\omega_c \vec{E}_{0\perp} e^{-i\omega_c t} + (d\vec{v}_\perp/dt) \times \vec{B}] \quad (3.41)$$

and using (3.40) to eliminate $d\vec{v}_\perp/dt$ in (3.41), we obtain, after rearranging the terms,

$$\frac{d^2\vec{v}_\perp}{dt^2} + \omega_c^2\vec{v}_\perp = -i\omega_c(q/m)\vec{E}_{0\perp}e^{-i\omega_c t} - (qB/m)(q/m)(\vec{B} \times \vec{E}_{0\perp})e^{-i\omega_c t} \quad (3.42)$$

The solution of this inhomogeneous differential equation is given by the sum of the solution of the homogeneous equation, plus a particular integral of the inhomogeneous equation. The solution of the homogeneous equation

$$d^2\vec{v}_\perp/dt^2 + \omega_c^2\vec{v}_\perp = 0 \quad (3.43)$$

is just the cyclotron motion described previously by the velocity $\vec{v}_m$. A particular integral of (3.42) is provided by the function

$$\vec{v}_\perp = \vec{A} t e^{-i\omega_c t} \quad (3.44)$$

To determine the vector $\vec{A}$ we differentiate (3.44) twice with respect to time

$$d\vec{v}_\perp/dt = \vec{A} (1 - i\omega_c t) e^{-i\omega_c t} \quad (3.45)$$

$$d^2\vec{v}_\perp/dt^2 = -2\vec{A} i\omega_c e^{-i\omega_c t} - \omega_c^2\vec{v}_\perp \quad (3.46)$$

Comparing (3.46) with (3.42) we see that (3.44) satisfies (3.42) provided we take

$$\vec{A} = (q/2m)\vec{E}_{0\perp} - (i/2\omega_c)(qB/m)(q/m)(\vec{B} \times \vec{E}_{0\perp}) \quad (3.47)$$

Therefore, the complete solution of (3.42) is

$$\vec{v}_\perp(t) = \vec{v}_m(t) + (q/2m)(\vec{E}_{0\perp} \mp i\vec{B} \times \vec{E}_{0\perp})t e^{-i\omega_c t} \quad (3.48)$$

where we have replaced qB/m by $\pm\omega_c$, the upper and lower signs corresponding, respectively, to positive and negative charges. Using (3.17) and (3.18), which define the right and left circularly polarized components of the electric field, respectively, we can write (3.48) for a *positively* charged particle ($q > 0$) as

$$\vec{v}_\perp(t) = \vec{v}_m(t) + (q/m)\vec{E}_{0L} t e^{-i\omega_c t} \quad (3.49)$$

and for a *negatively* charged particle ($q < 0$) as

$$\vec{v}_\perp(t) = \vec{v}_m(t) + (q/m) \vec{E}_{oR} t e^{-i\omega_c t} \quad (3.50)$$

Hence, the particle velocity increases indefinitely with time. Note that the expression $(q/m) \vec{E}_{oR,L}$ represents a constant acceleration. A positive charge resonates with $\vec{E}_L$, and a negative one with $\vec{E}_R$. The particle moves in circles of ever increasing radii, with its velocity increasing continually during this spiral motion at the expense of the energy of the electric field. A typical resonant spiral for an electron is shown in Fig. 2. This phenomenon can be used as a method of increasing the particle speed and hence the kinetic temperature of the plasma through collisions. This method is known as *radio frequency heating* of the plasma by *cyclotron resonance*.

4. TIME-VARYING MAGNETIC FIELD AND SPACE VARYING ELECTRIC FIELD

From Maxwell equations it is seen that a time-varying magnetic field is also accompanied by a space-varying electric field. The $\vec{E}$ field associated with the time-dependent $\vec{B}$ field is given by

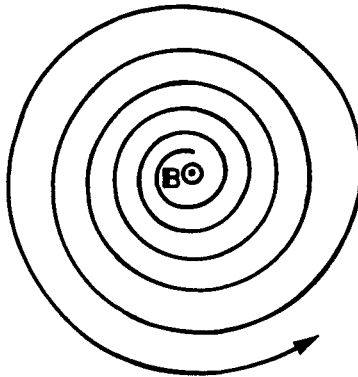


Fig. 2 - Outward spiral motion of an electron, in the plane normal to the magnetic field, under cyclotron resonance.

$$\vec{\nabla} \times \vec{E} = - \partial \vec{B} / \partial t \quad (4.1)$$

We shall assume that the fractional change in the magnetic field, in a time interval of the order of the cyclotron period, is very small.

4.1 Equation of motion and adiabatic invariants

Consider the magnetic field in the z direction and suppose that it is spatially uniform and increases with time within the circular orbit of the particle. From Faraday's law, an electric field is induced along the path of the particle orbit (Fig. 3) which accelerates the particle, with the result that the orbit is no longer a circle. However, since the time variation of the $\vec{B}$ field is small, the azimuthal component of the electric field $\vec{E}_\theta$ is also small, and the orbit will be nearly a circle.

Expressing (4.1) in cylindrical coordinates, taking $\vec{B} = B\hat{z}$, and considering $\vec{E} = \hat{\theta} E_\theta(r)$, we find

$$\frac{1}{r} \frac{\partial}{\partial r} (r E_\theta) = - \frac{\partial B}{\partial t} \quad (4.2)$$

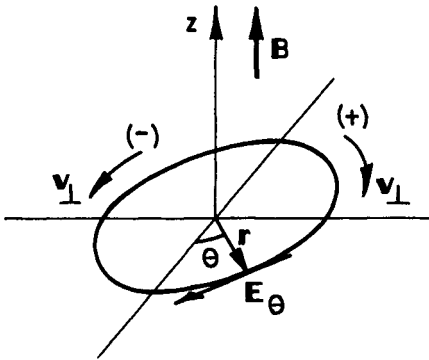


Fig. 3 - Azimuthal electric field $\vec{E}_\theta$ induced by a time-varying magnetic field. The magnetic field inside the orbit is uniform, parallel to the z axis, and increases slowly with time.

Integrating with respect to r , noting that $\partial B/\partial t$ can be taken outside the integral sign since B is a slowly varying function, yields

$$\int_0^r \partial/\partial r' (r' E_\theta) dr' = - \partial B/\partial t \int_0^r r' dr' \quad (4.3)$$

from which we obtain the induced electric field as

$$E_\theta = - (r/2) \partial B/\partial t \quad (4.4)$$

Further, since $\hat{\theta} = -\hat{r} \times \hat{z}$, (4.4) can be written in vector form as

$$\vec{E}_\theta = \vec{r} \times (\partial \vec{B}/\partial t)/2 \quad (4.5)$$

Using this result in the Lorentz force equation, we obtain for the equation of motion, after some rearrangement,

$$d\vec{v}/dt = (\partial \vec{\omega}_c/\partial t) \times \vec{r}/2 + \vec{\omega}_c \times \vec{v} \quad (4.6)$$

Instead of solving (4.6) directly, we shall determine a relation between the radius vector r and the time rate of change of $\vec{B}$, by calculating the change in the particle transverse kinetic energy over one gyration period, which results from the action of the induced electric field. Since the force acting on the particle due to the electric field is $q \vec{E}_\theta$, the increase in the transverse kinetic energy over one gyration period is given by

$$\delta(mv_\perp^2/2) = q \oint \vec{E}_\theta \cdot d\vec{r} \quad (4.7)$$

where $d\vec{r}$ denotes an element of path along the particle trajectory, so that $\vec{v}_\perp = d\vec{r}/dt$. Since the field changes very slowly, we can calculate the line integral in (4.7) as if the orbit were closed. Using Stoke's theorem, we replace the line integral by a surface integral over the unperturbed orbit,

$$\begin{aligned} \delta(mv_\perp^2/2) &= q \int_S (\vec{v} \times \vec{E}_\theta) \cdot d\vec{S} \\ &= - q \int_S (\partial \vec{B}/\partial t) \cdot d\vec{S} \end{aligned} \quad (4.8)$$

where use was made of (4.1). Here $\vec{S}$ denotes the surface enclosed by the cyclotron orbit, its direction being such that $\vec{B} \cdot d\vec{S} < 0$ for ions, and

$\vec{B} \cdot d\vec{S} > 0$ for electrons, in view of the diamagnetic character of a plasma. Thus (4.8) becomes

$$\delta(mv_{\perp}^2/2) = |q| (\partial B/\partial t) \pi r_C^2 \quad (4.9)$$

Now, the change in the magnetic field over one gyration period, $(2\pi/\omega_C)$, is

$$\delta B = (\partial B/\partial t)(2\pi/\omega_C) \quad (4.10)$$

and using the relations $r_C^2 = v_{\perp}^2/\omega_C^2$ and $\omega_C = |q| B/m$, (4.9) can be rewritten as

$$\delta(mv_{\perp}^2/2) = (mv_{\perp}^2/2B) \delta B = |\vec{m}| \delta B \quad (4.11)$$

where the quantity $|\vec{m}| = mv_{\perp}^2/2B$ is the orbital magnetic moment of the charged particle. Now, since the left-hand side of (4.11) is $\delta(|\vec{m}|B)$, we obtain

$$\delta|\vec{m}| = 0 \quad (4.12)$$

This result shows that the *magnetic moment* is *invariant* in slowly varying magnetic fields for which $(\partial B/\partial t)(2\pi/\omega_C) \ll B$.

From the constancy of the magnetic moment, we can easily verify that

$$B r_C^2 = \text{constant} \quad (4.13)$$

Therefore, as the magnetic field increases, the radius of gyration decreases, as shown in Fig. 4. Further, since the magnetic flux Φ_m through a Larmor orbit is given by

$$\Phi_m = BS = B\pi r_C^2 \quad (4.14)$$

it is clear that the *magnetic flux* is also an *adiabatic invariant*. Hence, as the magnetic field strength increases, the radius of the orbit decreases in such a way that the particle always encircles the same number of magnetic flux lines.

When the time variation of the magnetic field is not spatially uniform within the circular orbit of the gyrating particle, but if it occurs in an unsymmetrical way, then the induced electric field acting on the charged particle can considerably modify its orbit from the one shown in Fig. 4.

In the most general cases the particle orbit can be extremely complicated. In order to obtain a general idea of what may happen, let us consider the simple case of a magnetic field varying in time with cylindrical symmetry over a region of radius R , which is much larger than the cyclotron radius r_c , as shown in Fig. 5. The azimuthal component of the induced electric field, $\vec{E}_\theta$, at the point P (see Fig. 5) is given, from (4.5), by

$$\vec{E}_\theta = \vec{R} \times (\partial \vec{B} / \partial t) / 2 \quad (4.15)$$

A charged particle located at the point P is now acted upon by crossed electric and magnetic fields, resulting in a drift velocity given by

$$\vec{v}_E = (\vec{E}_\theta \times \vec{B}) / B^2 = (\vec{R} \times \partial \vec{B} / \partial t) \times \vec{B} / 2B^2 \quad (4.16)$$

Since $\vec{B}$ is in the z direction (normal to the vector $\vec{R}$) (4.16) gives

$$\vec{v}_E = - \vec{R} (\partial B / \partial t) / 2B \quad (4.17)$$

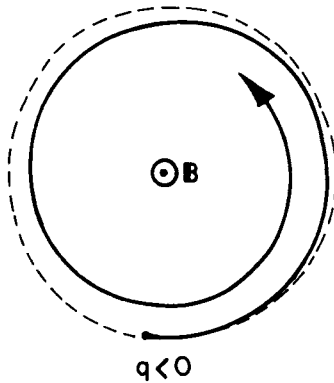


Fig. 4 - Motion of an electron in a time-varying magnetic field. The field is uniform and increases with time inside the particle orbit.

Therefore, the guiding center of the particle drifts radially inward with the drift velocity $\vec{v}_E$ given in (4.17). As the particle drifts radially inward, its radius of gyration decreases in such a way that the flux encircled by the gyrating particle remains constant (Fig. 5). Since the density of the magnetic flux lines increases as the magnetic field strength increases, this radial particle drift can be pictured as a radially inward motion of the magnetic flux lines with the velocity $\vec{v}_E$ given in (4.17), with the guiding center attached to a given flux line.

4.2 Magnetic heating of a plasma

The adiabatic invariance of the orbital magnetic moment of the charged particle ($|\vec{m}| = mv_{\perp}^2/2B = \text{constant}$) implies that when the magnetic field is increased, the

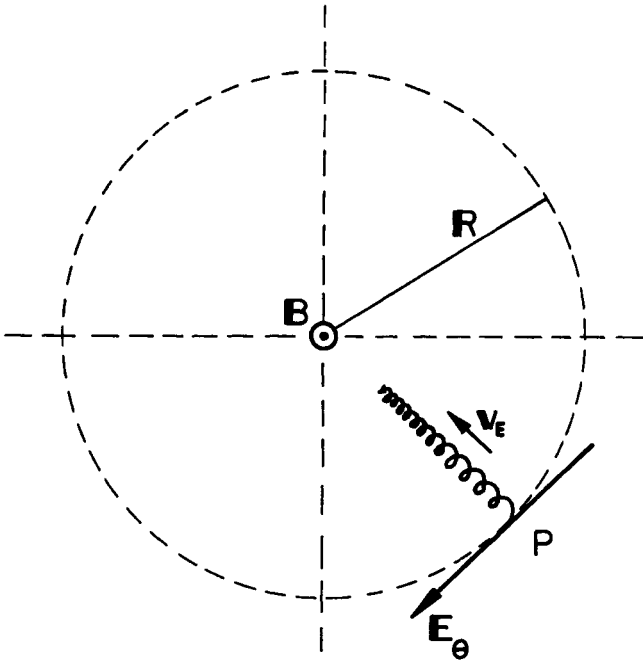


Fig. 5 - Motion of a negatively charged particle in an increasing magnetic field (directed out of the page) with cylindrical symmetry over a region of radius R much larger than the Larmor radius r_c .

particle transverse kinetic energy ($W_{\perp} = mv_{\perp}^2/2$) increases linearly with B . Further, since the magnetic flux encircled by the gyrating particle is also constant, as the magnetic flux density increases, the magnetic flux tube contracts and the particle guiding center moves radially inward, accompanying the radial displacement of the magnetic field lines, as if it was frozen in the field lines. As a consequence, the increase in the magnetic flux density causes the charged particles in a plasma to approach each other, resulting in a *magnetic compression*. In the present case of two-dimensional compression, since the increase in the number density (n) of the particles is proportional to the cross-sectional area πr_c^2 , it follows that under magnetic compression in two dimensions n increases linearly with B . Similarly, when B decreases, n also decreases, resulting in a *magnetic decompression*. Thus, for two-dimensional compression we have $W_{\perp} \propto B \propto n$.

This property is used as a method of plasma heating, known as *magnetic pumping*, which consists of periodic magnetic compressions and decompressions of the plasma. The compression and the decompression take place in a time interval that is very large compared to the Larmor period but, at the same time, is very small compared to the relaxation time necessary for the achievement of thermal equilibrium. For the present case of two-dimensional compression-decompression, obtained by varying the axial magnetic field, only the velocity and the energy in the two directions normal to the magnetic field are changed. In order to be able to heat the plasma in a compression-decompression cycle, it is necessary to transfer part of the energy increase in $W_{\perp}$, obtained by compression, to the energy $W_{\parallel}$ which is unaffected by decompression. This energy transfer is brought about by collisional interactions, which promotes equipartition of energy, therefore increasing the particle thermal kinetic energy at the expense of the electric field energy.

Thus, in a complete cycle consisting of a compression, a relaxation time, and a decompression, part of the energy increase in $W_{\perp}$ achieved during compression is transferred to the degree of freedom parallel to B , as a result of collisions, causing an increase in $W_{\parallel}$ which is not affected during the decompression, whereas $W_{\perp}$ is decreased. Therefore, by a periodic repetition of these cycles of adiabatic compression-decompression, the energy of the plasma and therefore its temperature is increased.

5. SUMMARY OF GUIDING CENTER DRIFTS AND CURRENT DENSITIES

5.1 Guiding center drifts

$$\text{Electric field:} \quad \vec{v}_E = (\vec{E} \times \vec{B})/B^2 \quad (2.5.15)$$

$$\text{Gravitational field:} \quad \vec{v}_g = m(\vec{g} \times \vec{B})/(qB^2) \quad (2.6.3)$$

$$\text{General force:} \quad \vec{v}_F = (\vec{F} \times \vec{B})/(qB^2) \quad (2.6.2)$$

$$\text{Gradient of } \vec{B}: \quad \vec{v}_G = -|\vec{m}|(\vec{\nabla}B) \times \vec{B}/(qB^2) \quad (3.5.1)$$

$$\text{Curvature of } \vec{B}: \quad \vec{v}_C = -mv_{||}^2[(\vec{B} \cdot \vec{\nabla})\vec{B}] \times \vec{B}/(qB^4) \quad (3.7.9)$$

$$\text{Gradient-curvature of } \vec{B}: \quad \vec{v}_{GC} = -m(v_{||}^2 + v_{\perp}^2/2)(\vec{\nabla}B^2/2) \times \vec{B}/(qB^4) \quad (3.8.2)$$

(in vacuum field)

$$\text{Polarization:} \quad \vec{v}_p = m(\partial \vec{E}_{\perp}/\partial t)/(qB^2) \quad (2.11)$$

5.2 Current densities

$$\text{Magnetization:} \quad \vec{J}_M = \vec{\nabla} \times \vec{M} \quad (2.4.40)$$

$$\text{Gravitational:} \quad \vec{J}_g = \rho_m \vec{g} \times \vec{B}/B^2 \quad (2.6.5)$$

$$\text{Gradient of } \vec{B}: \quad \vec{J}_G = -[(1/\Delta V) \sum_i |\vec{m}_i|] (\vec{\nabla}B) \times \vec{B}/B^2 \quad (3.5.3)$$

$$\text{Curvature of } \vec{B}: \quad \vec{J}_C = -2 [(1/\Delta V) \sum_i W_{i||}] [\vec{B} \cdot \vec{\nabla})\vec{B}] \times \vec{B}/B^2 \quad (3.7.12)$$

$$\text{Polarization:} \quad \vec{J}_p = \rho_m (\partial \vec{E}_{\perp}/\partial t)/B^2 \quad (2.12)$$

PROBLEMS

4.1 - With reference to a magnetic field pointing along the z axis ($\vec{B} = B_0 \hat{z}$), describe the type of polarization of the following electric field:

$$\vec{E} = E_0 (\hat{x} \cos \omega t - \hat{y} \sin \omega t)$$

Make a drawing which shows the orientation of $\vec{E}$ for the instants $t = 0$, $t = \pi/2\omega$ and $t = \pi/\omega$. How can you represent this electric field in complex notation?

4.2 - Describe, in a semiquantitative way, the motion of an electron in the presence of a constant magnetic field

$$\vec{B} = B_0 \hat{z}$$

and a time-varying electric field given by

$$\vec{E} = (E_0/2) (\hat{x} + i \hat{y}) \exp(-i\omega_c t)$$

where E_0 and B_0 are positive constants and $\omega_c = e B_0/m_e$. What type of polarization has this electric field?

4.3 - Solve the equation of motion to determine the transient response of a charged particle in the presence of a spatially uniform a.c. electric field $\vec{E}(t) = \hat{x} E_0 \sin(\omega t)$, that is switched on at $t = 0$. Assume that initially, at $t = 0$, the particle is at rest at the origin. Make a plot of the particle trajectory and velocity as a function of time.

4.4 - Consider an electron acted by a constant and uniform magnetic field $\vec{B} = B_0 \hat{z}$, and a uniform but time-varying electric field $\vec{E} = \hat{y} E_{y0} \sin(\omega t)$. Assume that the initial conditions are such that the motion takes place in the x-y plane and that at $t = 0$ the electron is at rest ($\vec{v}_0 = 0$) at the origin.
(a) Show that the orbit of the electron is given by

$$x(t) = -(eE_{y0}/m)[(\omega/\omega_c)(\cos\omega_c t - 1) - (\omega_c/\omega)(\cos\omega t - 1)]/(\omega^2 - \omega_c^2)$$

$$y(t) = -(eE_{y0}/m)[(\omega/\omega_c)\sin(\omega_c t) - \sin(\omega t)]/(\omega^2 - \omega_c^2)$$

(b) In the low-frequency limit, $\omega \ll \omega_c$, show that the electron orbits at the angular frequency ω around an ellipse which has its major axis perpendicular to the electric field. Determine the ratio of the minor to the major axis of the ellipse.

(c) In the high-frequency limit, $\omega \gg \omega_c$, show that the electron moves in a circle at the cyclotron frequency ω_c .

4.5 - Integrate (3.49) and (3.50) to determine the particle trajectory in the plane normal to B and sketch the path of the particle for $q > 0$ and $q < 0$.

4.6 - Consider the motion of an electron in the presence of a uniform magnetostatic field $\vec{B} = B_0 \hat{z}$, and an electric field which oscillates in time at the electron cyclotron frequency ω_c , according to

$$\vec{E}(t) = E_0 [\hat{x} \cos(\omega_c t) + \hat{y} \sin(\omega_c t)]$$

(a) What type of polarization has this electric field?

(b) Obtain the following uncoupled differential equations satisfied by the velocity components $v_x(t)$ and $v_y(t)$:

$$d^2 v_x / dt^2 + \omega_c^2 v_x = 2(eE_0/m) \omega_c \sin(\omega_c t)$$

$$d^2 v_y / dt^2 + \omega_c^2 v_y = -2(eE_0/m) \omega_c \cos(\omega_c t)$$

(c) Assume that, at $t = 0$, the electron is located at the origin of the coordinate system, with zero velocity. Neglect the time-varying part of $\vec{B}$. Show that the electron velocity is given by

$$v_x(t) = -(eE_0/m)t \cos(\omega_c t)$$

$$v_y(t) = -(eE_0/m)t \sin(\omega_c t)$$

(d) Show that the electron trajectory is given by

$$x(t) = -(eE_0/m) [(1/\omega_c^2) \cos(\omega_c t) + (t/\omega_c) \sin(\omega_c t) - 1/\omega_c^2]$$

$$y(t) = -(eE_0/m) [(1/\omega_c^2) \sin(\omega_c t) - (t/\omega_c) \cos(\omega_c t)]$$

4.7 - Solve the equation of motion to determine the velocity and the trajectory of an electron in the presence of a uniform magnetostatic field $\vec{B} = B_0 \hat{z}$, and an oscillating electric field given by

$$\vec{E}(t) = \hat{x} E_x \sin(\omega t) + \hat{z} E_z \cos(\omega t)$$

Consider the same assumptions and initial conditions as in Problem 4.6.

4.8 - Consider the motion of an electron in a spatially uniform magnetic field $\vec{B} = B_z \hat{z}$, such that B_z has a slow time variation given by

$$B_z(t) = B_0(1 - \alpha t)$$

where B_0 and α are positive constants, and $|\alpha t| \ll 1$. Assume the following initial conditions: $\vec{r}(0) = (r_c, 0, 0)$ and $\dot{\vec{r}}(0) \equiv \vec{v}(0) = (0, v_{z0}, 0)$, where r_c is the Larmor radius, $v_{z0} = \omega_c r_c$, and $\omega_c = |q| B_0/m$.

(a) Write the equation of motion, considering the Lorentz force, and solve it by a perturbation technique including only terms up to the first order in the small parameter α . Show that the particle velocity is given by

$$v_x(t) = -\omega_c r_c \sin(\omega_c t) + \frac{1}{2} \alpha r_c \omega_c t [\sin(\omega_c t) + \omega_c t \cos(\omega_c t)]$$

$$v_y(t) = \omega_c r_c \cos(\omega_c t) + \frac{1}{2} \alpha r_c \omega_c t [-\cos(\omega_c t) + \omega_c t \sin(\omega_c t)]$$

(b) Show that the particle orbit is given by

$$x(t) = r_c [(1 + \alpha t/2) \cos(\omega_c t) + (\alpha/2\omega_c)(\omega_c^2 t^2 - 1) \sin(\omega_c t)]$$

$$y(t) = r_c [(1 + \alpha t/2) \sin(\omega_c t) - (\alpha/2\omega_c)(\omega_c^2 t^2 - 1) \cos(\omega_c t) - \alpha/2\omega_c]$$

$$z(t) = v_{z0} t$$

(c) Determine the orbital magnetic moment and verify its adiabatic invariance, retaining only terms up to the first order in α .

4.9 - Consider the motion of a charged particle in a spatially uniform magnetic field which varies slowly in time as compared to the cyclotron period of the particle.

(a) Show that the equation of motion can be written in vector form as

$$d\vec{v}(t)/dt = \vec{\omega}_c(t) \times \vec{v}(t) + [\partial \vec{\omega}_c(t)/\partial t] \times \vec{r}(t)/2$$

where $\vec{\omega}_c(t) = -q \vec{B}(t)/m$.

(b) Considering that $\vec{B}(t) = \hat{z} B_0 f(t)$, where B_0 is constant, obtain the following equations for the motion of the particle in the plane normal to $\vec{B}$,

$$\frac{d^2 x(t)}{dt^2} = \omega_c \left[\frac{dy(t)}{dt} f(t) + \frac{1}{2} y(t) \frac{df(t)}{dt} \right]$$

$$\frac{d^2 y(t)}{dt^2} = -\omega_c \left[\frac{dx(t)}{dt} f(t) + \frac{1}{2} x(t) \frac{df(t)}{dt} \right]$$

where $\omega_c = |q| B_0/m$.

(c) Define a complex variable $u(t) = x(t) + iy(t)$, and a function $\xi(t)$ by

$$\xi(t) = u(t) \exp \left[(i\omega_c/2) \int_0^t f(t') dt' \right]$$

and show that the equation satisfied by $\xi(t)$ is

$$\frac{d^2 \xi(t)}{dt^2} + (\omega_c^2/4) f^2(t) \xi(t) = 0$$

(d) If $\xi_1(t)$ and $\xi_2(t)$ are two linearly independent solutions of this equation, subjected to the initial conditions

$$\xi_1(0) = 0 \quad d\xi_1/dt|_{t=0} = 1$$

$$\xi_2(0) = 1 \quad d\xi_2/dt|_{t=0} = 0$$

show that the solution for $u(t)$ can be written as

$$u(t) = \exp \left[-(i\omega_c/2) \int_0^t f(t') dt' \right] \{ u_0 \xi_2(t) + \xi_1(t) [\dot{u}_0 + (i\omega_c/2) f(0) u_0] \}$$

where u_0 and $\dot{u}_0$ represent the initial position and velocity respectively.

(e) Considering now that the particle is initially ($t = 0$) at the origin and moving with velocity v_0 along the negative y axis, that is, $u_0 = 0$ and $\dot{u}_0 = -iv_0$, show that

$$u(t) = \exp \left[-(i\omega_c/2) \int_0^t f(t') dt' \right] [-iv_0 \xi_1(t)]$$

and, consequently,

$$x(t) = v_0 \xi_1(t) \sin[(\omega_c/2) \int_0^t f(t') dt']$$

$$y(t) = -v_0 \xi_1(t) \cos[(\omega_c/2) \int_0^t f(t') dt']$$

4.10 - (a) Assume that $f(t)$, in Problem 4.9, is given by $\exp(-\alpha t)$. Show that, in this case, $\xi(t)$ satisfies the Bessel equation of zero order,

$$\frac{d^2 \xi(\tau)}{d\tau^2} + (1/\tau) \frac{d\xi(\tau)}{d\tau} + \xi(\tau) = 0$$

where $\tau \equiv (\omega_c/2\alpha) \exp(-\alpha t)$. Determine the two solutions of this equation which satisfy the initial conditions stated in Problem 4.9, and interpret them physically.

(b) Considering now that $f(t) = (1 - \alpha t)$, solve the equation for $\xi(t)$ in Problem 4.9, in a power series in α , and determine the particle trajectory to order α . Show that the ratio $(v_x^2 + v_y^2)/B(t)$ has no terms of order α , thus verifying the adiabatic invariance of the magnetic moment. Compare these results with those of Problem 4.8.

4.11 - For an electron, with initial velocity $v_0 \hat{x}$ and initial position $x_0 \hat{x}$, acted upon by an electric field $\vec{E} = \hat{x} E \cos(kx - \omega t)$, show that its velocity is given by

$$v(t) = v_0 - (eE/m_e) \int_0^t \cos(kx - \omega t') dt'$$

Using a perturbation approach, in which to lowest order $\vec{E} = 0$, show that

$$v(t) = v_0 - [(eE/m_e)/(kv_0 - \omega)] \{ \sin[kx_0 + (kv_0 - \omega)t] - \sin(kx_0) \}$$

Notice that the velocity perturbation will be large only when v_0 is close to the phase velocity ω/k .

4.12 - Using the Maxwell equation (1.5.3), and (3.34) which defines the plasma conductivity dyad $\vec{\sigma}$, and considering the time variation indicated in (3.1), show that

$$\vec{\nabla} \times \vec{B} = -i\omega\mu_0 \vec{\epsilon} \cdot \vec{E}$$

where $\bar{\epsilon}$ is the plasma electric permittivity dyad given by

$$\bar{\epsilon} = \epsilon_0 [\bar{1} + (i/\omega\epsilon_0) \bar{\sigma}]$$

where $\bar{1}$ denotes the unit dyad which in Cartesian coordinates can be written as

$$\bar{1} = \hat{x}\hat{x} + \hat{y}\hat{y} + \hat{z}\hat{z}$$